H → WW + charm

Analysis status

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2022postEE · 26.7 fb^{-1} · expected UL 1034

Sections

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Dedicated decks: `ctag-sf-deck.pdf` (34 slides) · `negrw-deck.pdf` (57 slides)

1 · Analysis and selection

Signal and final state

H → WW → 2ℓ2ν, produced with a charm quark.

Final state: **opposite-sign eμ** + MET + **≥1 c-tagged jet**

object	selection
muons	tight ID + iso, $p_T > 10$, $\|\eta\| < 2.4$
electrons	wp80iso, $p_T > 10$, $\|\eta\| < 2.5$
ll pair	OS, $p_T > 20$ / 10, $m_{ll} > 12$
c-jets	$p_T > 20$, PNet medium CvL/CvB

> eμ removes the Z peak by construction — the dominant background is **real $t\bar{t}$**.

Data: ReReco `22Sep2023`, all eras.

2 · MVA-defined regions

Six classes, one network

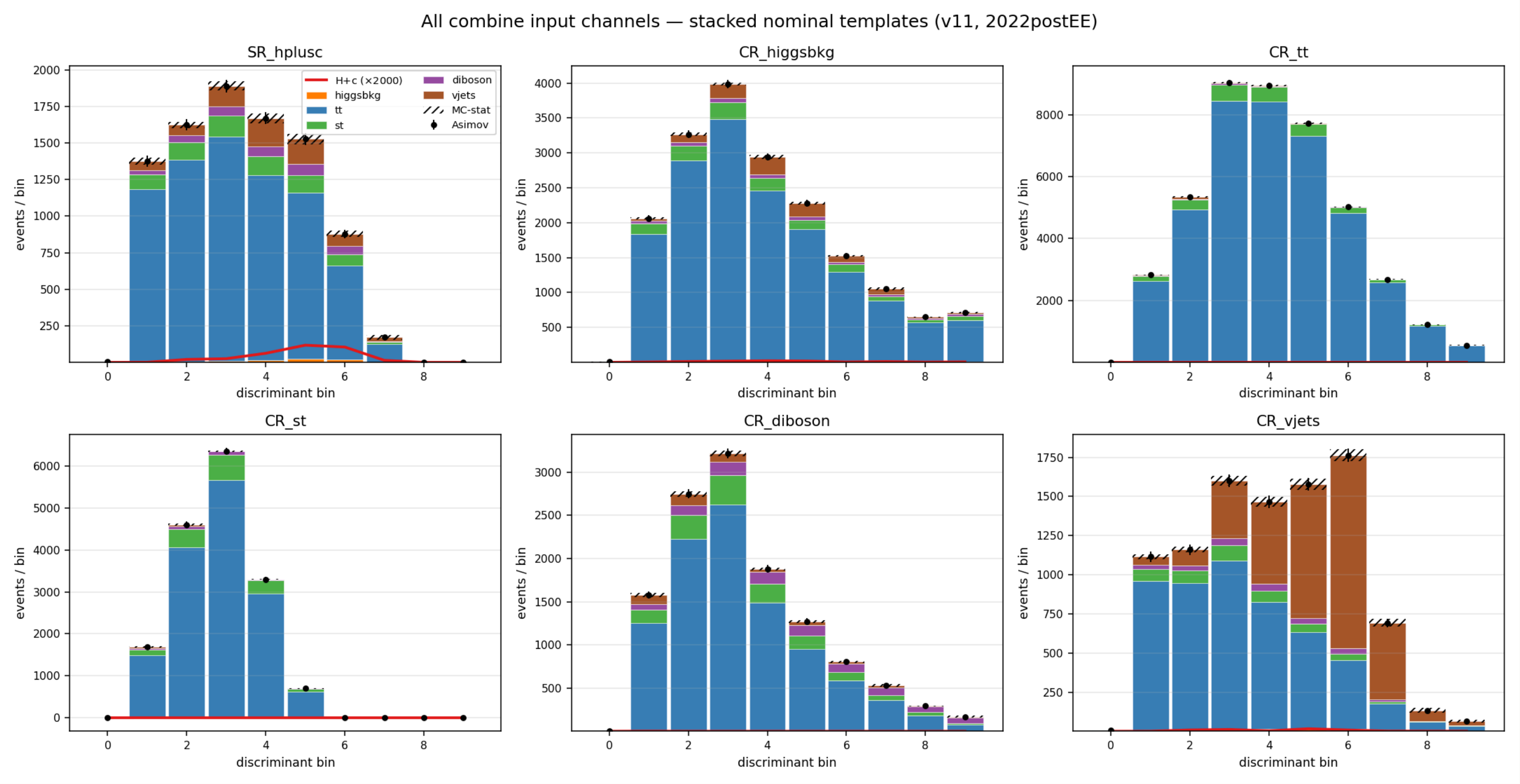
`[hplusc, higgsbkg, tt, st, diboson, vjets]`

argmax defines the region · the winning score is the discriminant

region	events	dominant
SR_hplusc	20,664	tt 82%
CR_tt	44,199	**tt 94%**
CR_vjets	9,214	**vjets 43%**
CR_higgsbkg / st / diboson	6.6k–20k	tt-rich

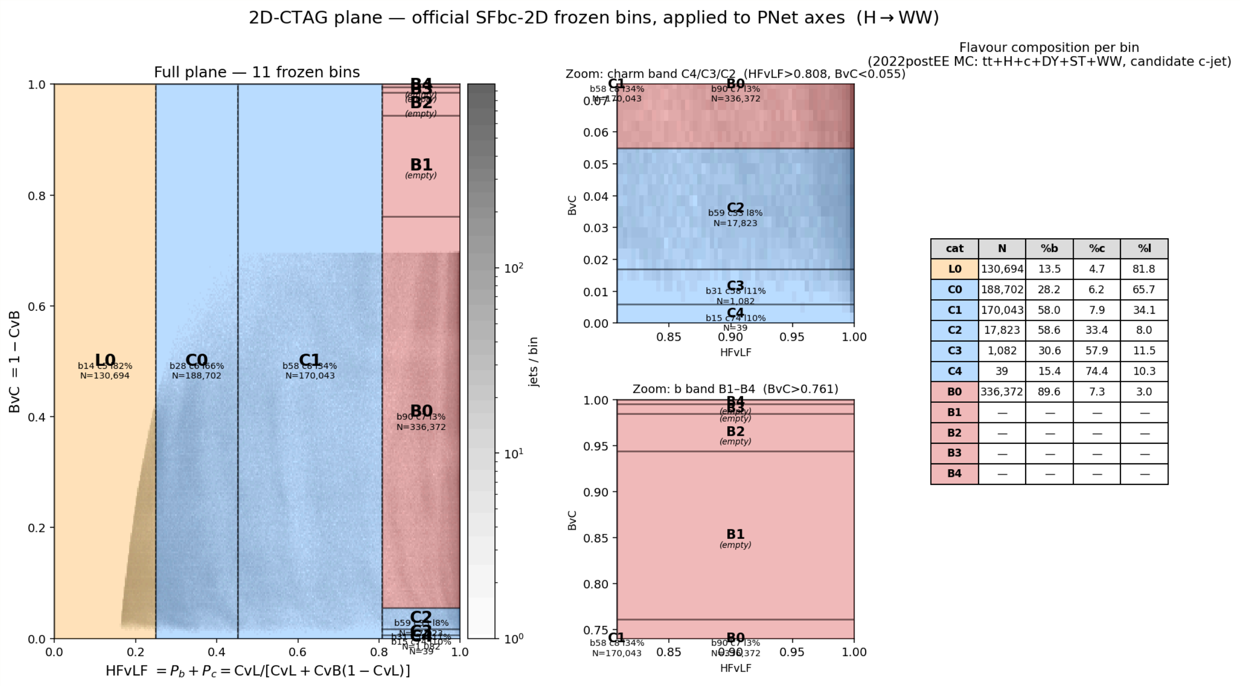
> CR_tt at **94% purity** pins the free-floating `rate_tt`, which covers 82% of the SR.

Templates in all six regions



3 · c-tagging

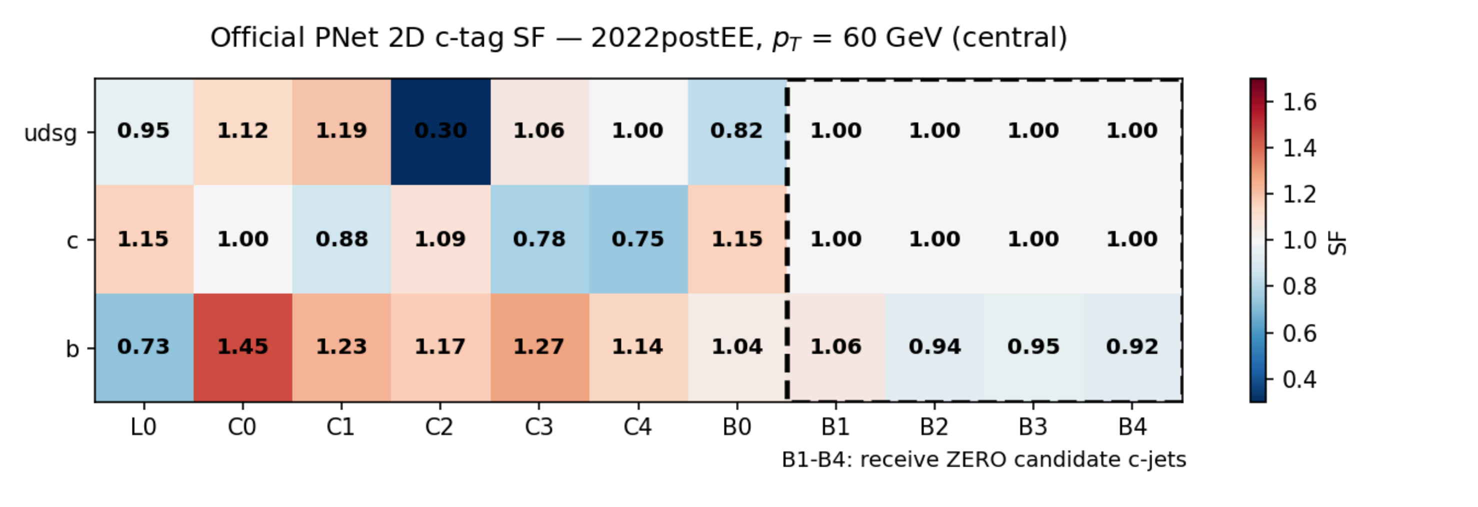
The 2D plane



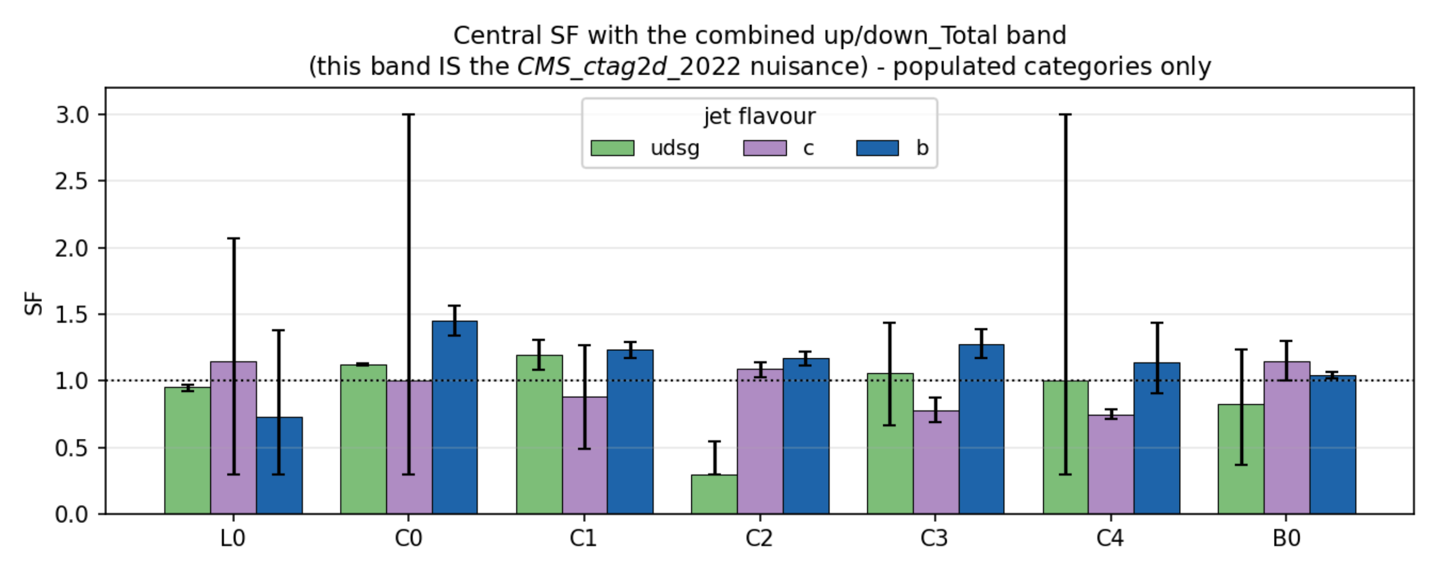
cat	N	%b	%c	%l
L0	130,694	13.5	4.7	81.8
C0	188,702	28.2	6.2	65.7
C1	170,043	58.0	7.9	34.1
C2	17,823	58.6	33.4	8.0
C3	1,082	30.6	57.9	11.5
C4	39	15.4	74.4	10.3
B0	336,372	89.6	7.3	3.0
B1	—	—	—	—
B2	—	—	—	—
B3	—	—	—	—
B4	—	—	—	—

11 categories `L0, C0–C4, B0–B4` spanning the whole CvL/CvB plane.

The scale factor matrix



The uncertainty band is the nuisance



What the calibration costs

configuration	limit
no SF	1150
+ `CMS_ctag2d_2022`	**1164**

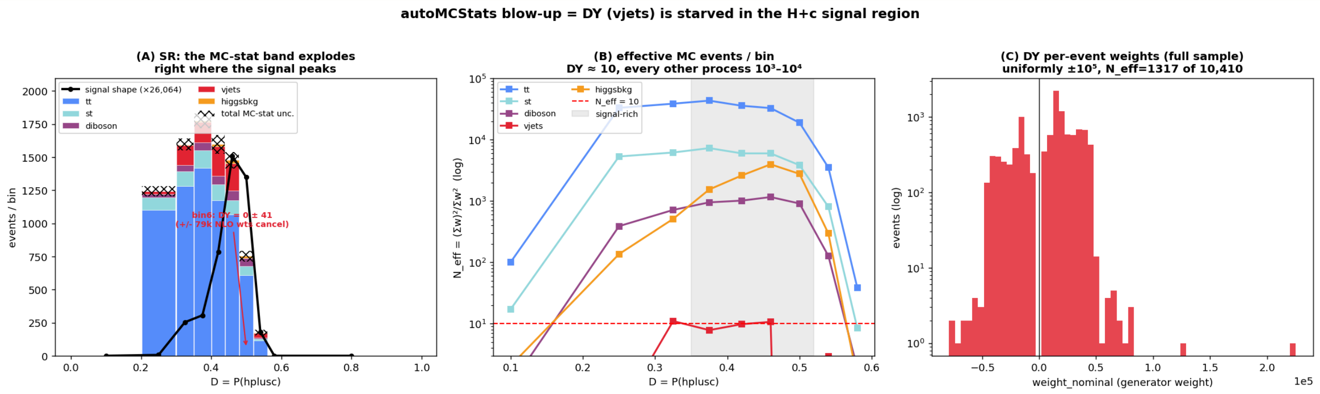
Applying the calibration **worsens** the limit by 14 units — as it must.

A scale factor adds a nuisance. The point is correctness.

Currently **one nuisance** covering the whole plane. Decorrelation is pending.

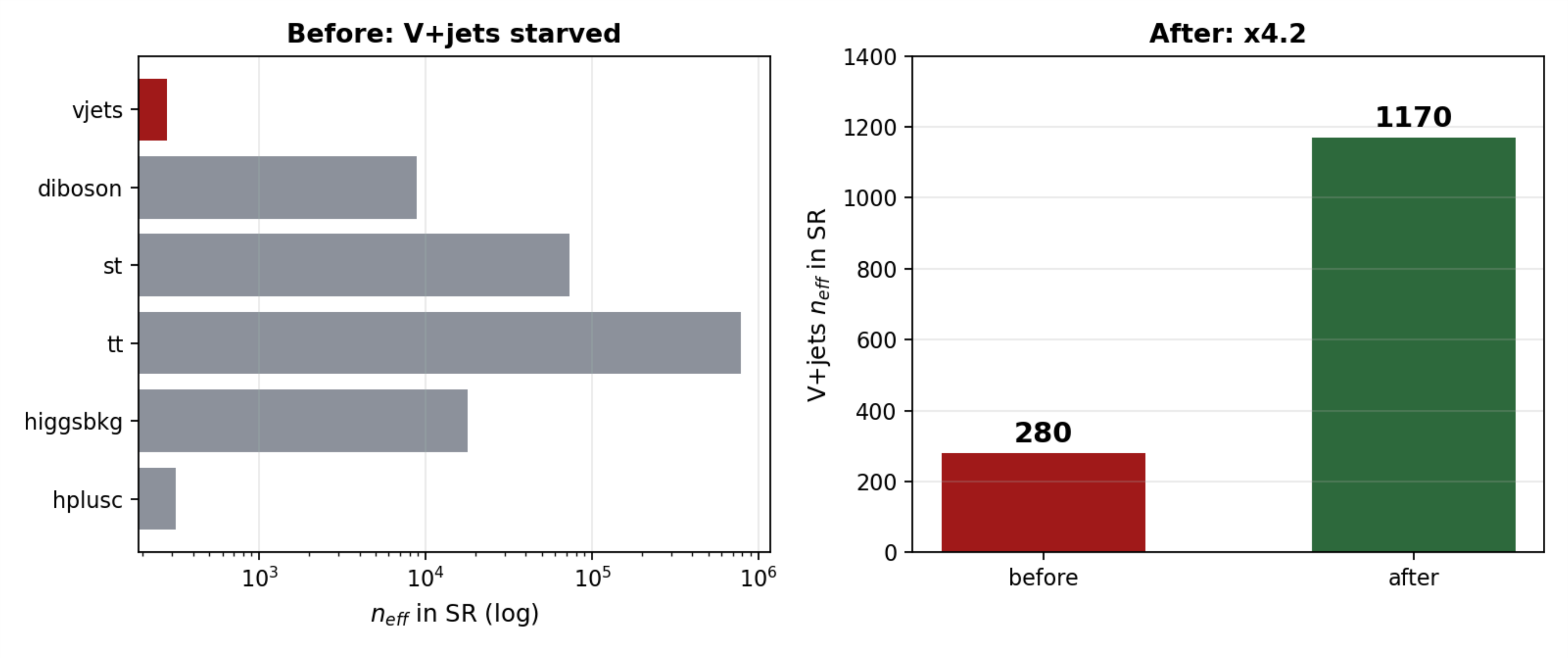
4 · Negative-weight reweighting

The problem



aMC@NLO weights are **signed** — the yield is a cancellation.

V+jets was starved exactly under the signal



$n_{eff} = (\sum w)^2 / \sum w^2$ — every other background sits at ≤1.1%.

The method

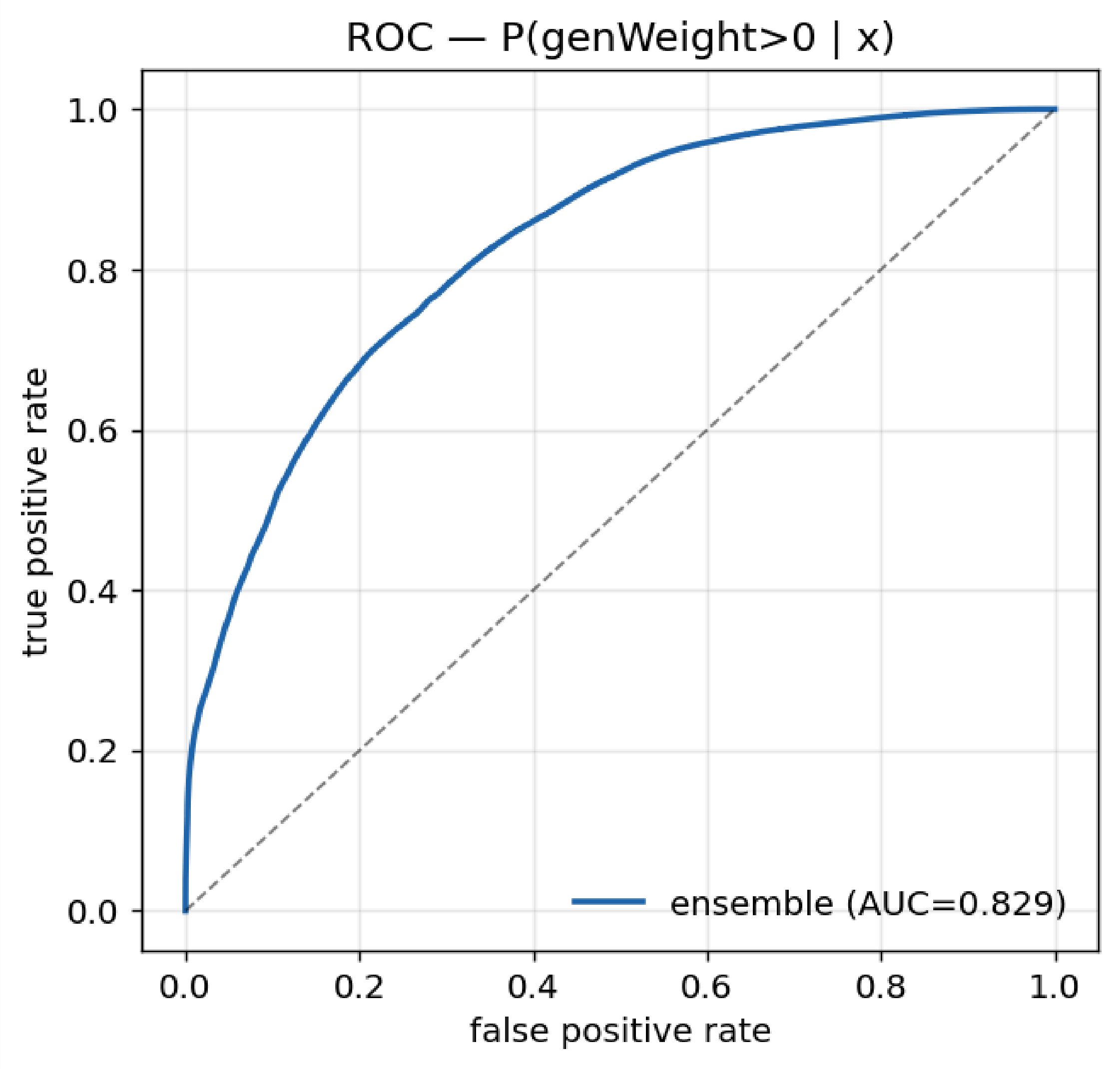
$$w \;\to\; |w| \cdot g(x) \cdot \mathrm{renorm}, \qquad g(x) = 2P_+(x) - 1$$

An algebraic identity, not an approximation — yield-preserving by construction.

- **generator-level features only** — P_+ is a property of the generator
- **train loose, infer tight**, disjoint by construction from the eμ SR
- same phase space, disjoint events → interpolation, never extrapolation

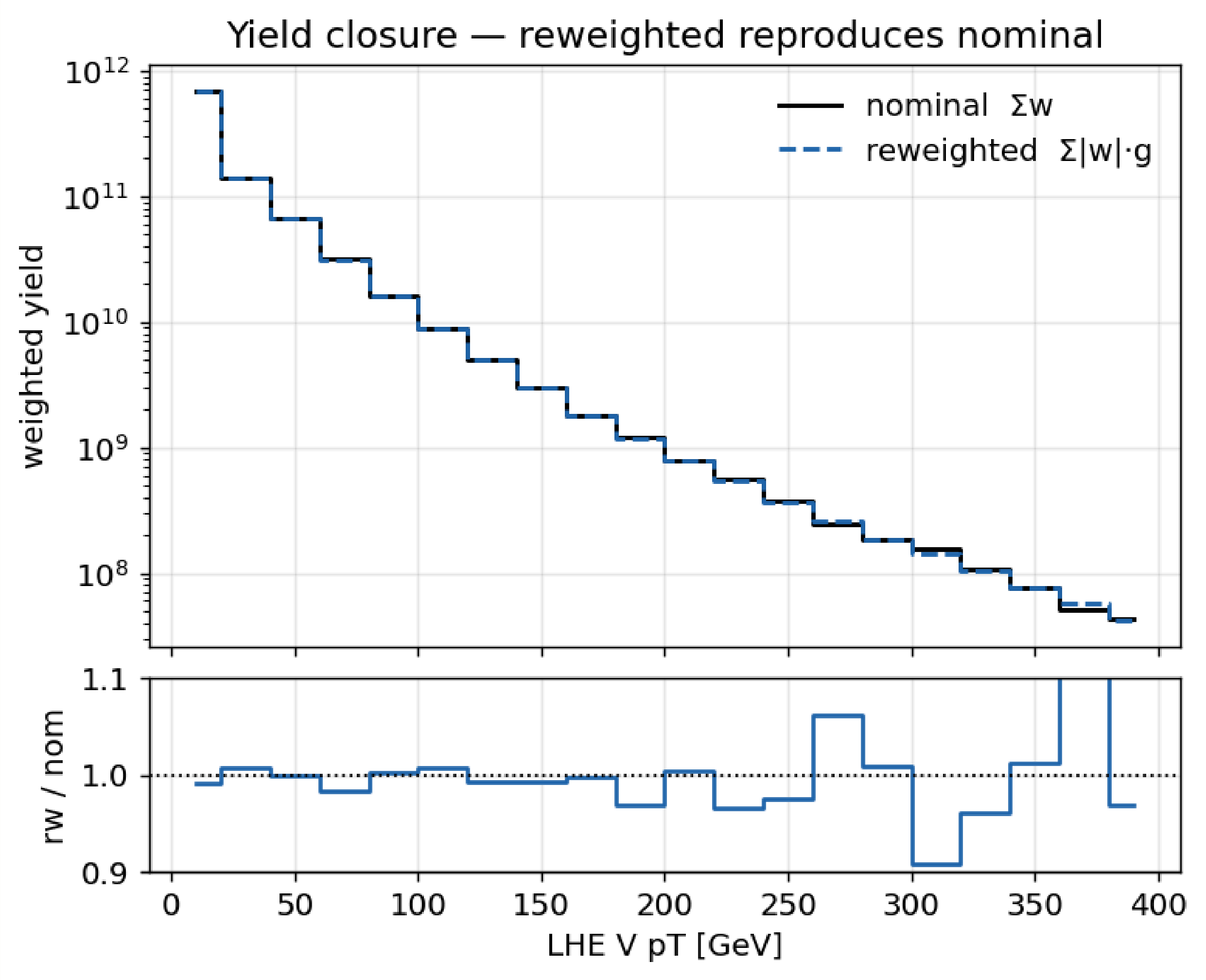
arXiv:2510.16217

Classifier performance



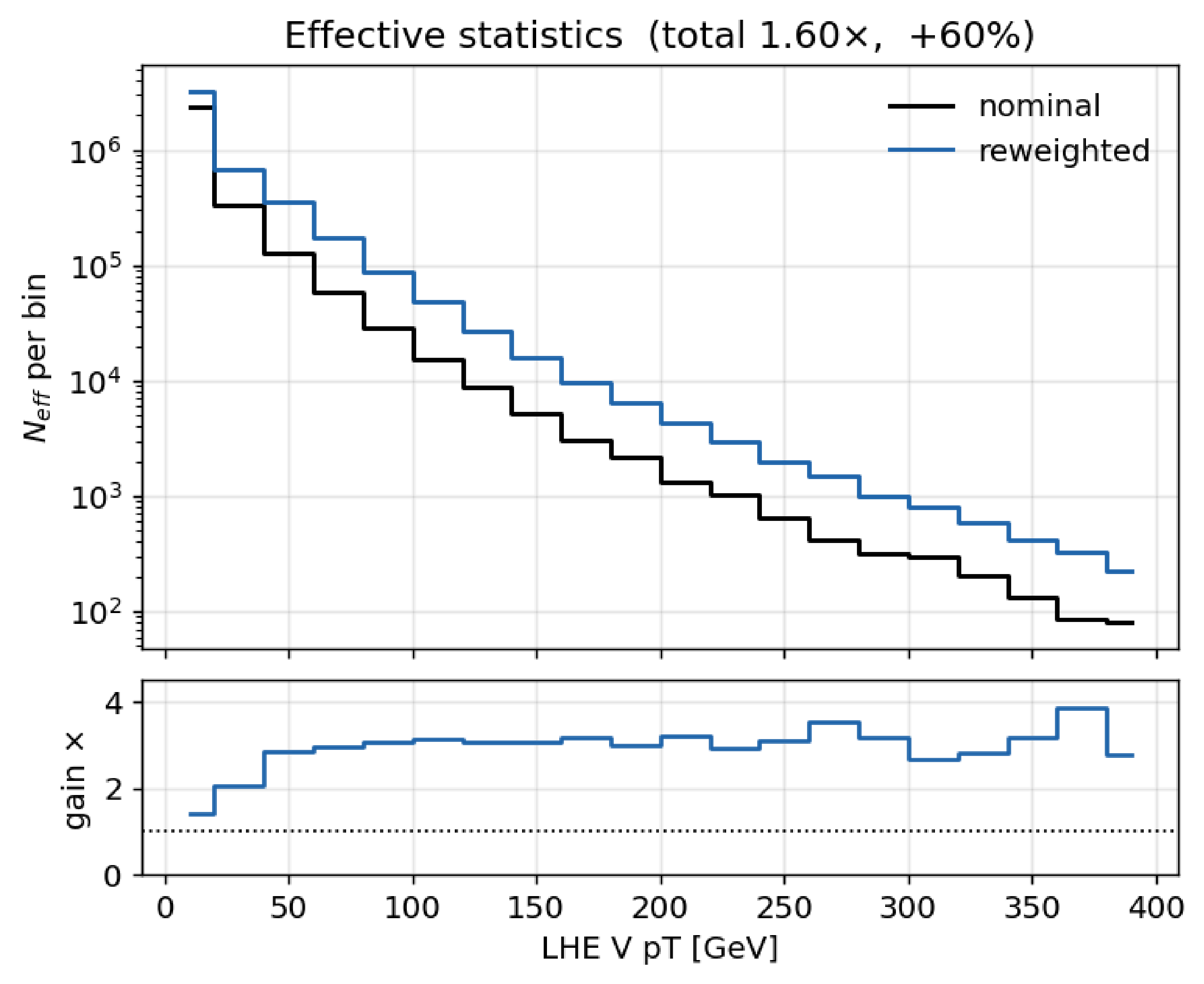
Ensemble **AUC 0.829** on an intrinsically stochastic target.

Closure



Training-region closure **0.994** on 9.8M events.

The gain



Method uncertainty `CMS_negrw_vjets` profiles to **0.0%** — negligible.

5 · W+jets jet-binned samples

Inclusive → jet-binned

AN-23-102 rejects the inclusive NLO sample: *large negative-weight fraction.*

sample	xsec (pb)	events	neg-w
0J	55,760	678M	10%
1J	9,529	523M	26%
2J	3,532	345M	35%
sum	**68,821**	**1,546M**	
inclusive	*67,710*	*282M*	*16%*

Cross sections from XSDB · sum within **+1.6%** of the inclusive.

Result

	before	after
expected UL	1160	**1034**
V+jets n_{eff} (SR)	280	**1170**
V+jets stat. error	5.98%	**2.92%**
V+jets rate	1508	1523

The rate barely moved while the statistical error halved — the signature of a statistics gain, not a normalisation change.

Was the gain really W+jets?

`tt` (+4.8%) and `st` (+3.8%) also moved between the two cards — both were re-merged from scratch. Checked per-process:

process	n_{eff} ratio
vjets (SR)	**4.18×**
vjets (CR_st)	**9.89×**
vjets (CR_tt)	**5.86×**
tt, st, diboson, higgsbkg	0.87 – 1.13×

The n_eff gain is confined to V+jets. Other processes move by ≤13% in *both* directions — re-merge jitter, not a systematic shift. Both cards were also re-verified independently: **1160** and **1034**.

6 · Systematics

The full inventory

22 shape · 9 lnN · 1 rateParam · autoMCStats

shape (weight)	shape (object shift)
`pileup`	`CMS_scale_j_2022` (JES)
`ps_isr`, `ps_fsr`	`CMS_res_j_2022` (JER)
`scalevar_muR`, `_muF`, `_muR_muF`	`CMS_scale_e_2022`, `CMS_res_e_2022`
`lhe_pdf`, `lhe_alphaS`	`CMS_scale_m_2022`, `CMS_res_m_2022`
`muon_id`, `muon_iso`	
`electron_id`, `electron_reco` ×3	**lnN**
`CMS_ctag2d_2022`	`lumi_13p6TeV`, `xsec_st/diboson/vjets/higgsbkg`
`CMS_negrw_vjets`	`BR_HtoWW`, `BR_Htautau`, `xsec_hplusc_4FS_5FS`

Plus **`rate_tt`** (free rateParam, tt from data) and **autoMCStats** (threshold 10).

Object shifts need full reprocessing

JES/JER and lepton scale/resolution **change the MVA score**, so they cannot be applied as an event weight.

Each is a **separate parquet tree**, re-run through the full selection *and* re-scored by the network — 12 shifted directories.

> **This is the trap that once cost 500 units.** A partial inference run scored only 19 of 57 directories and left object-shift templates frozen at nominal; the limit read 1676 instead of 1185. Inference coverage is now verified after every rebuild.

What we fixed — 1 · top-p_T reweighting

Our first implementation was **wrong** and was corrected against `hh2bbww` (the Hamburg framework behind AN-24-091).

	what we had	what we now apply
form	data-based exponential	**theory-based (NNLO/NLO)**
Run 3	none	**× $(0.991 + 7.5\text{e-}5 \cdot p_T)$**
nominal	left at 1.0	**reweighted**
p_T cap	invented 500 GeV	**none** (form is well-behaved)

```
sf_run2 = 0.103*exp(-0.0118*pT) - 0.000134*pT + 0.973
sf      = (0.991 + 0.000075*pT) * sf_run2
weight  = sqrt(prod(sf))        # over the two gen tops
down    = 1.0                   # "no correction" is the variation
```

Our original coefficients were real — but belonged to the *other*, data-driven variant.

What we fixed — 2 · Higgs heavy-flavour

The original **flat lnN on merged `higgsbkg` was mis-scoped**: ggH is only **13.1%** of that group (VBF 29.0%, ggZH 23.3%, ZH 21.0%, WH 9.1%).

A flat lnN either over-penalises the other 87% (1.40) or must be watered down to an average (1.066) — and **cannot produce a shape effect at all**.

Replaced with a per-event weight, ported from HiggsDNA `Higgs_plus_HF_syst`:

```
num_HF_jets = ak.sum(genJets.hadronFlavour == 4, axis=-1)   # pT>25, |eta|<2.5
up   = where(num_HF_jets > 0, 1.5, 1.0)      # AN-23-102: 50% on ggH
down = where(num_HF_jets > 0, 0.5, 1.0)
```

> Keys on **gen-jet flavour**, so process grouping stops mattering — and it produces the shape a flat lnN structurally cannot.

What we fixed — 3 · MET unclustered energy

The `CorrectedMETFactory` path **never runs for Run 3 PuppiMET**, so the unclustered-energy shift was silently absent.

Now taken directly from the NanoAOD branches:

```
events.PuppiMET.ptUnclusteredUp / ptUnclusteredDown
events.PuppiMET.phiUnclusteredUp / phiUnclusteredDown
```

Independently confirmed against HiggsDNA `MET_syst_Unclustered` — same branches, same up/down ordering.

What we added

systematic	what was done
`CMS_negrw_vjets`	**new** — ensemble spread of the negative-weight reweighting
`CMS_ctag2d_2022`	**new** — 2D CvL/CvB SF applied natively in the processor
`electron_reco` ×3	**split** into p_T bins (<20, 20–75, >75 GeV)
`lhe_pdf`, `lhe_alphaS`	**added** — NNPDF replicas, MC2Hessian
`top_pt`	**rewritten** to the hh2bbww theory form
`higgs_plus_c`	**new** — per-event HF weight replacing the lnN
MET unclustered	**new** object shift

What we checked and deliberately excluded

item	decision
muon reco SF	**not applicable** — HiggsDNA exposes no reco key; hh2bbww registers `mu_id_sf` / `mu_iso_sf` and no `mu_reco_sf`. Two frameworks, same conclusion.
PU jet ID	absent from both frameworks for Run 3
UE / tune	requires **dedicated tune samples** — cannot be a weight
`hdamp`, `mtop`	sample-based tt modelling; absent from AN-23-102 Table 16 too

> `hdamp` and `mtop` are standard Run 3 tt modelling terms. Both need alternative samples. Recorded as a **documented decision**, not an oversight.

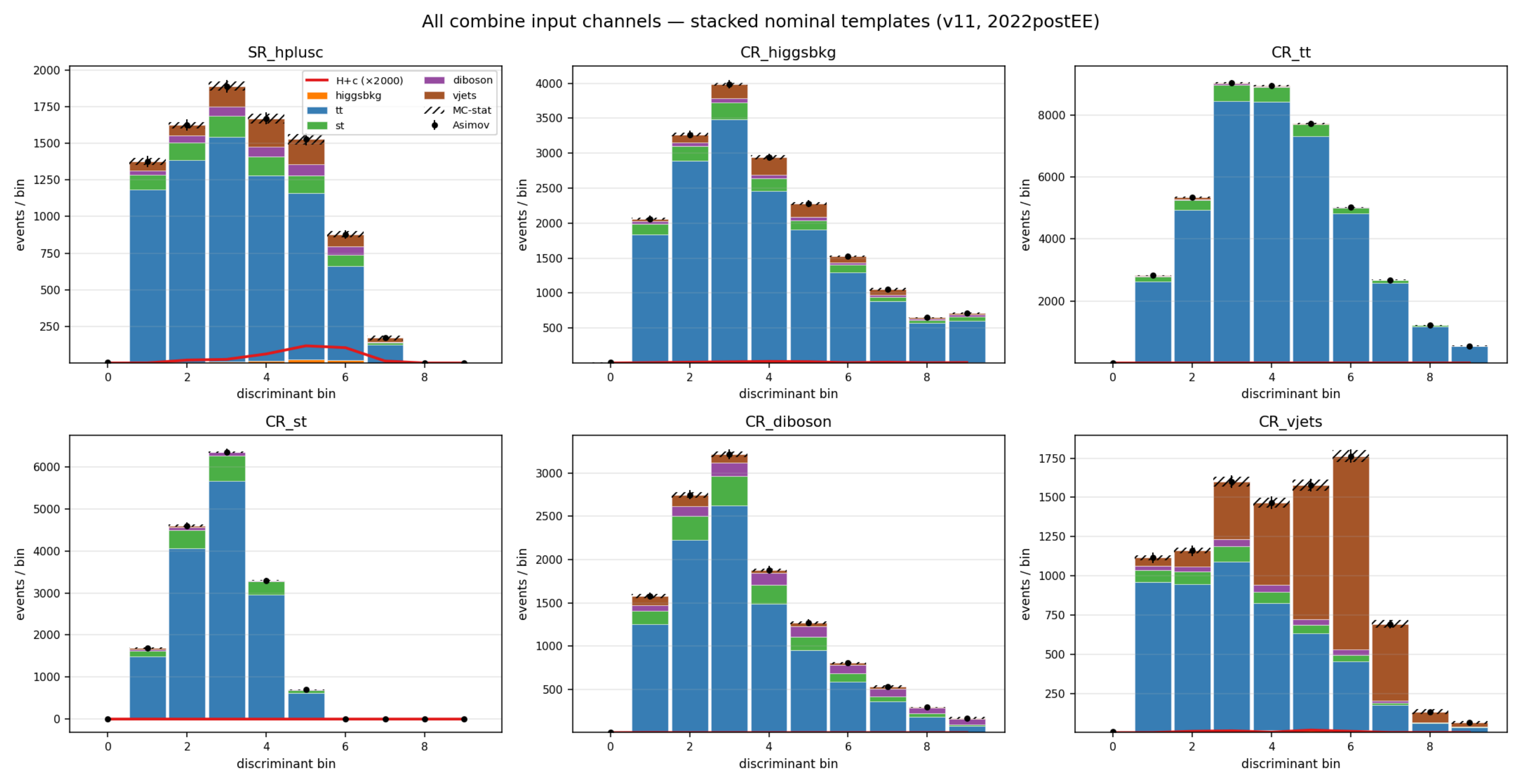
The result

	limit	fraction
full	**1034**	100%
statistical only	**641**	**62%**
systematics	—	**38%**

62% statistical · 38% systematic

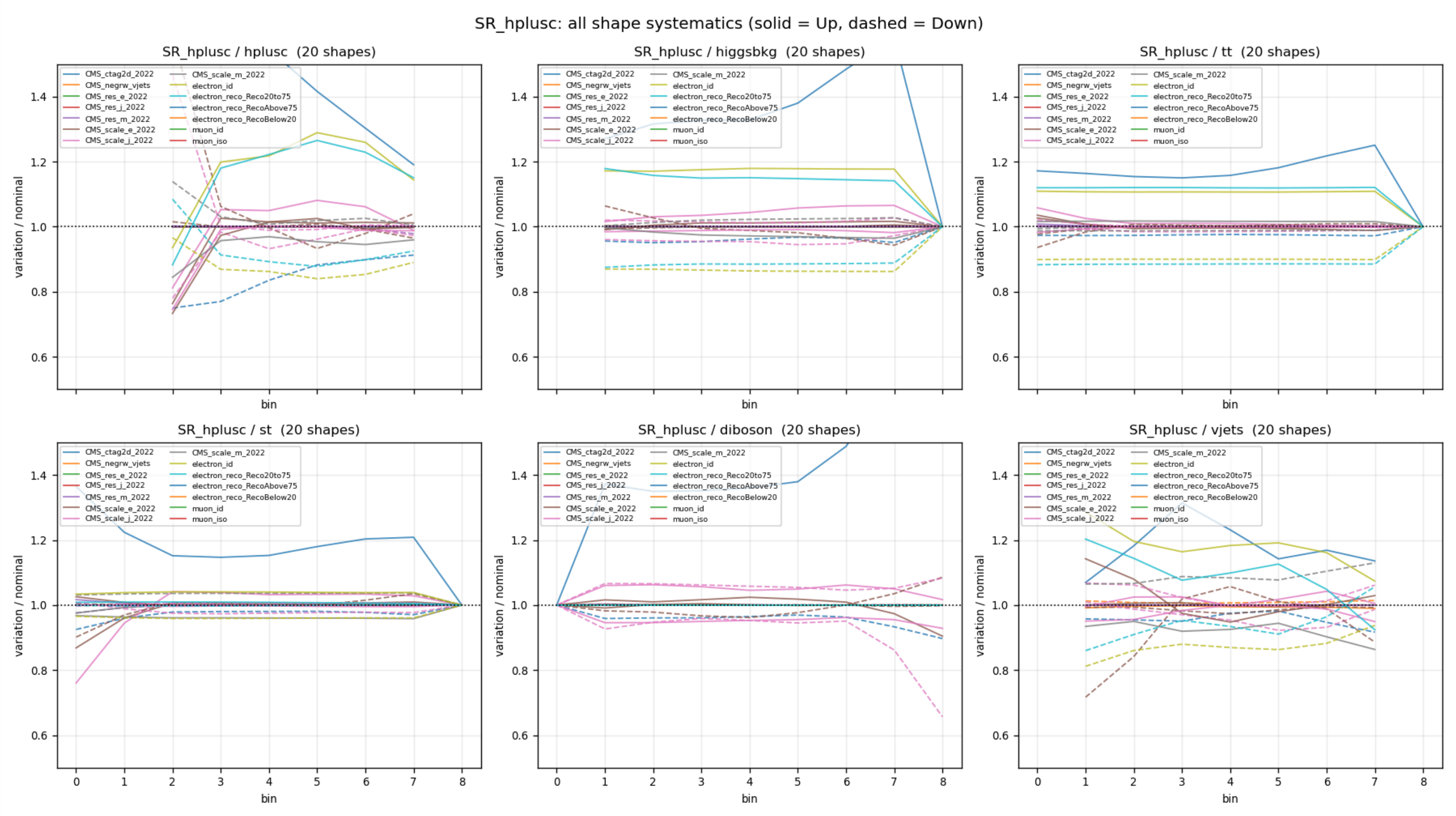
The 1034 number is **under investigation** — the systematic breakdown is not final and should not be quoted term-by-term.

Templates entering combine



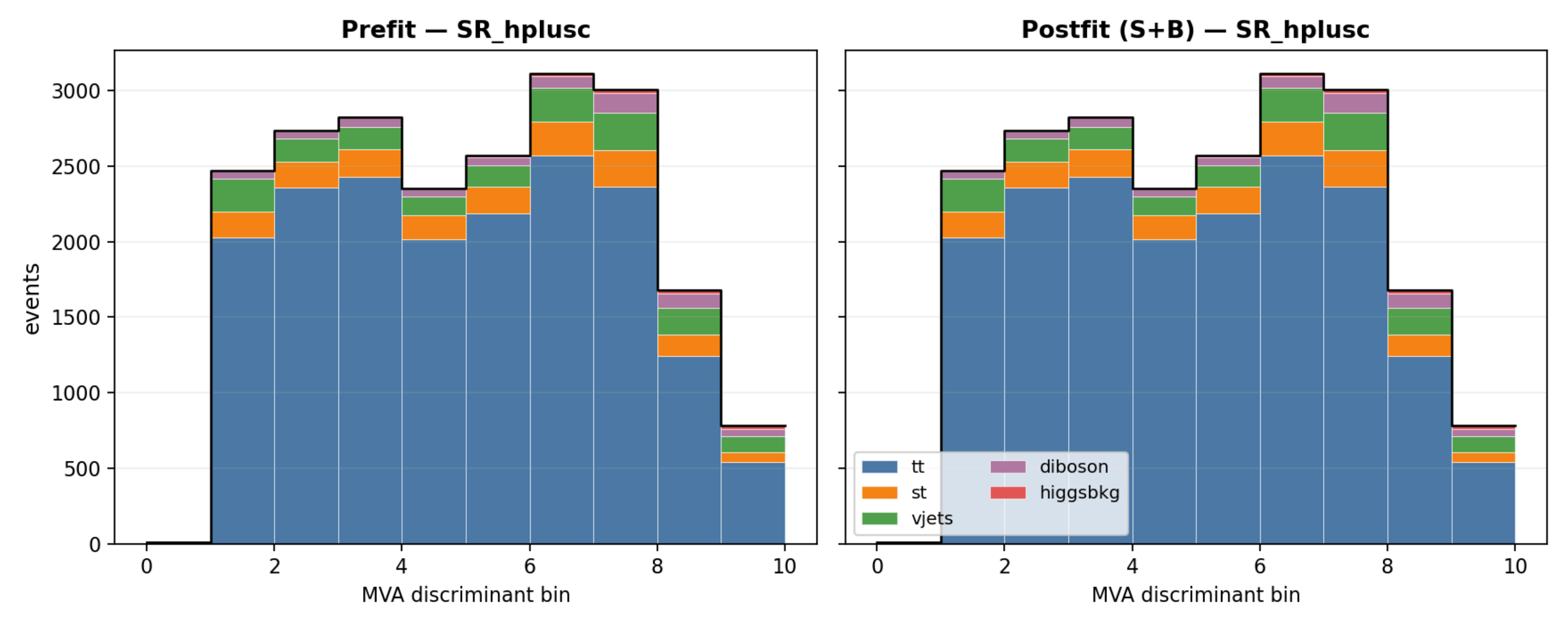
6 channels × 6 processes × 10 bins

Shape systematics in the signal region



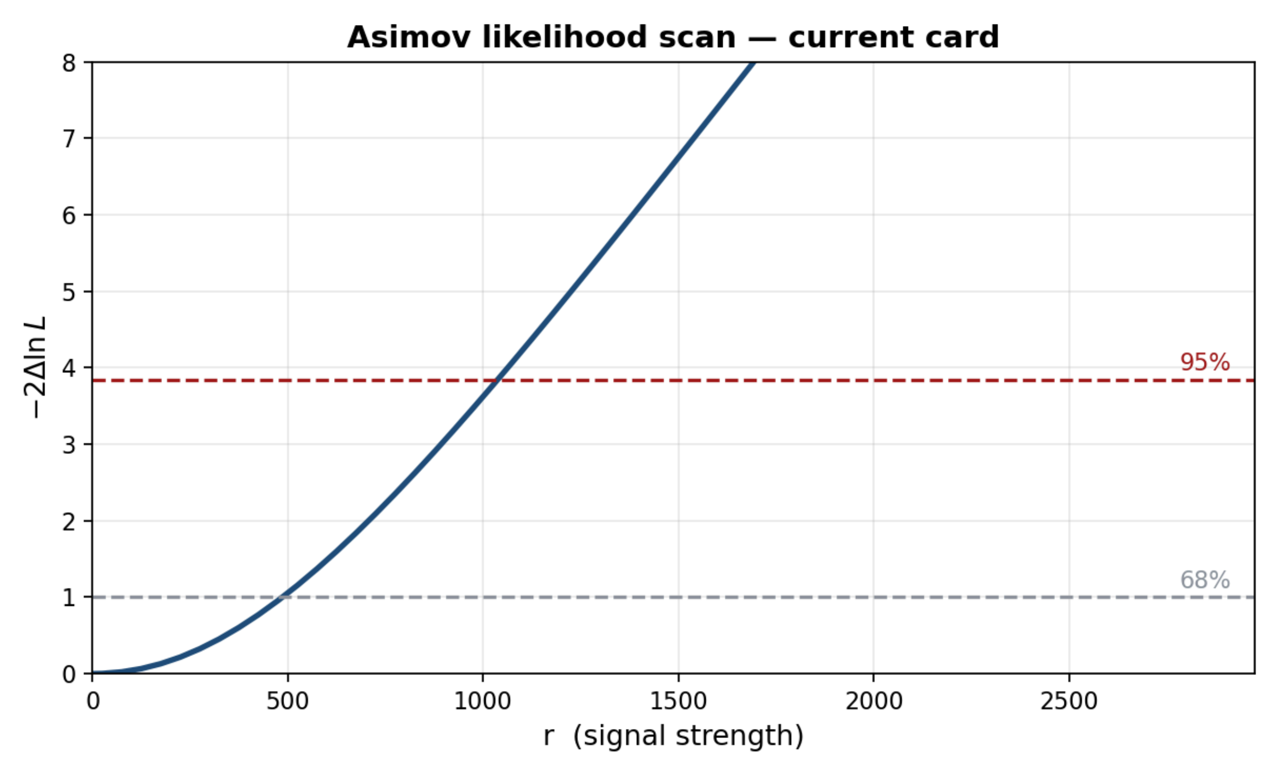
All shape nuisances, up (solid) vs down (dashed), ratio to nominal

Prefit vs postfit — signal region

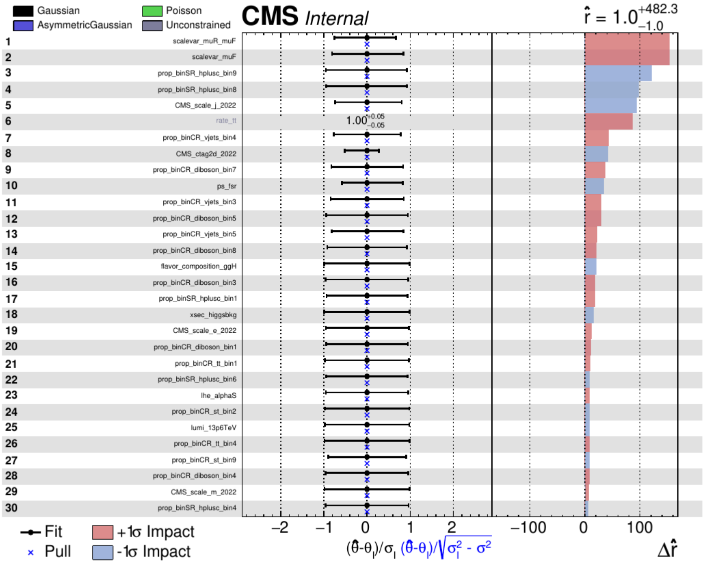


Asimov fit, r = 1 injected

Likelihood scan

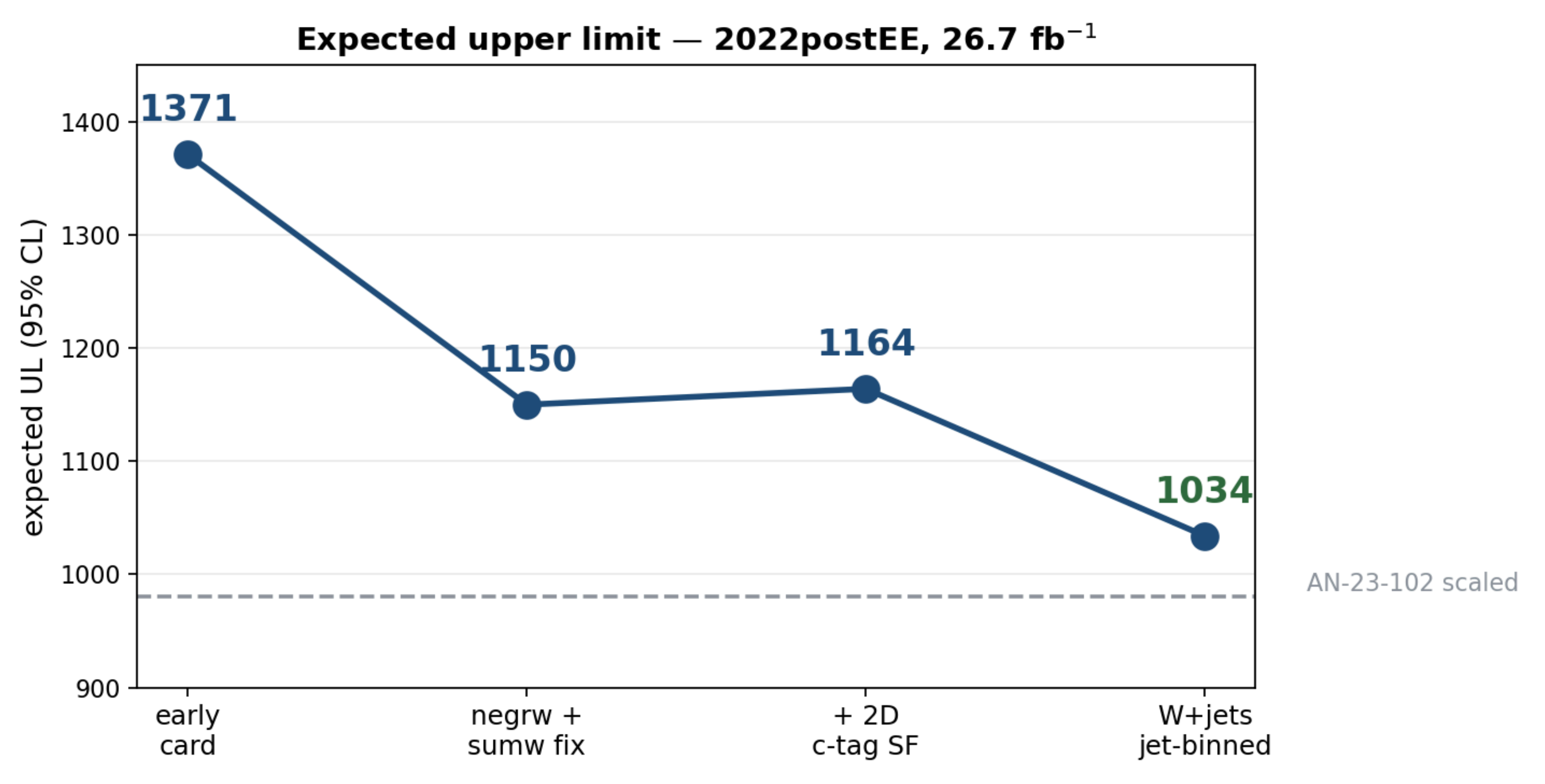


Nuisance impacts



7 · Status and outlook

The road so far



Where we stand

	limit
AN-23-102 scaled to 26.7 fb^{-1}	980
this analysis	**1034**
our statistical only	**641**

Within **~5%** of the published analysis scaled to our luminosity, on **5× less data**.

The systematic breakdown of the 1034 is **under investigation** — the full term-by-term comparison against AN-23-102 Table 17 is not yet settled.

What is still missing

item	status
4FS/5FS signal theory	placeholder 1.30 — needs re-derivation at 13.6 TeV
Trigger scale factors	implemented, **not yet enabled**
Higgs heavy-flavour (`higgs_plus_c`)	implemented, pending activation
top-p_T reweighting	implemented, pending activation
Dataset substitutions	W+jets p_T-binned (full AN stitch)
	DY → $Z\to\tau\tau$ filtered
	tt / single-top `-ext` samples
	`WZto3LNu`

Everything else in AN-23-102 Table 16 is **covered and in the card**:
22 shape + 9 lnN + `rate_tt` + autoMCStats.

Priorities

#	item	why
1	Re-derive `xsec_hplusc_4FS_5FS`	30% flat lnN on a statistics-starved signal
2	Enable trigger SFs	implemented, one flag
3	Activate `higgs_plus_c` + `top_pt`	proper HF and tt treatment
4	Add the missing datasets	more MC statistics
5	Decorrelate `CMS_ctag2d_2022`	currently one nuisance over the whole plane

Summary

Expected UL 1371 → 1034 · statistical-only **641**

V+jets n_{eff} **280 → 1170**

Within ~5% of the Run 2 analysis scaled to our luminosity

Remaining: 4FS/5FS re-derivation, trigger SFs, dataset substitutions